LAND REGISTRATION SYSTEM

Blockchain Integration — Enhanced with EHR-Inspired Architecture

Complete Team Reference Guide | 12 Members | 3 Organizations | 3 Orderers

🟡 Orderer Org	🔵 Org1	🟢 Org2	🔴 Org3
Members 1–3	Members 4–6	Members 7–9	Members 10–12
Consensus	Registration Dept	Revenue Dept	Collectorate

🔗 DigiLocker	🆔 DID	🌐 IPFS	🤖 AI/OCR
Doc Verification	Land Owner Identity ⭐	Document Storage ⭐	Doc Auto-read ⭐

⭐ = Enhancement inspired by EHR/Medi-Locker Architecture

1. What is Enhanced in This Project

The original Land Registration System is enhanced by borrowing 4 powerful ideas from the EHR (Medi-Locker) architecture and applying them specifically to land records.

Enhancement	Borrowed From EHR	How It Applies to Land Registration
🆔 DID — Decentralized Identity	EHR uses DID for each patient	Each land owner gets a unique DID on blockchain. Their property ownership is permanently linked to this DID. No identity fraud possible.
🌐 IPFS Storage	EHR stores encrypted health files on IPFS	Land documents (sale deeds, survey maps, encumbrance certificates) stored on IPFS as encrypted files. Only CID (fingerprint) stored on blockchain.
🤖 AI/OCR Pipeline	EHR uses FastAPI+PyTesseract+SpaCy to read medical docs	Land documents (old sale deeds, manual survey reports, paper certificates) are automatically read by OCR and converted to structured JSON for blockchain.
🔐 Consent / Access Control	EHR uses consent smart contracts for doctor access	Banks, lawyers, or buyers requesting to view land records must get owner's explicit approval on blockchain. Every access is logged permanently.

Before vs After Enhancement

Feature	Original System	Enhanced System ⭐
Owner Identity	X.509 Certificate only	X.509 Certificate + DID (permanent, tamper-proof blockchain identity)
Document Storage	Files uploaded to server folder	Files encrypted with AES-256 and stored on IPFS. Only CID on blockchain.
Document Reading	Manual upload of PDFs	OCR+NLP auto-reads scanned documents and converts to structured JSON
Document Verification	DigiLocker hash check	DigiLocker hash check + IPFS CID verification + AI confidence scoring
Record Access	Anyone with login can view	Owner must give explicit consent for each bank/lawyer/buyer to access records
Audit Trail	Basic blockchain history	Full access log: who accessed what, when, why — permanently on blockchain
Old Paper Documents	Not supported	OCR pipeline converts old physical documents to digital blockchain records

2. Enhanced Technologies — Simple Explanations

🆔 DID — Decentralized Identifier for Land Owners

⭐ **ENHANCEMENT (Inspired by EHR Architecture) — This feature is NEW, added to make Land Registration smarter and more secure**

In the EHR project, every patient gets a DID as their permanent health identity. We borrow this same idea for land owners.

Concept	Explanation
What is DID?	A unique permanent digital identity for each land owner generated on blockchain. Format: did:hf:org1:<unique-hash>. Cannot be duplicated or faked.
Why for land records?	Currently, land ownership uses paper documents that can be forged. A DID makes ownership cryptographically provable — nobody can claim to be the owner without the private key.
How it is created?	When a citizen registers on the portal, Fabric CA + chaincode generates a DID. This DID is linked to all their property records permanently.
What replaces?	Replaces fragile certificate-based identity with a self-sovereign, globally unique identifier that the owner controls.
Real example	did:hf:org1:a3f92bc1d04e5f678 → This is Rajesh Kumar's permanent land ownership identity on blockchain

🌐 IPFS — Document Storage for Land Records

⭐ **ENHANCEMENT (Inspired by EHR Architecture) — This feature is NEW, added to make Land Registration smarter and more secure**

In the EHR project, health records are encrypted and stored on IPFS. We apply the same to land documents.

Concept	Explanation
What is IPFS?	InterPlanetary File System — a global decentralized storage network. Files stored here get a unique CID (Content Identifier) that never breaks or expires.
Why for land records?	Land documents (sale deeds, maps, certificates) are large files. Storing them directly on blockchain is too expensive. IPFS stores the file; blockchain stores only the CID fingerprint.
How it works?	Document uploaded → AES-256 encrypted → uploaded to IPFS → IPFS returns CID → CID stored on blockchain linked to owner DID.
What is CID?	A unique hash fingerprint of the document. Example: QmXoypizjW3WknFiJnKLwHCnL72vedxjQkDDP1mXWo6uco. If file changes even 1 byte, CID completely changes — tamper detection.
Documents stored on IPFS	Sale Deed PDF, Survey Map image, Encumbrance Certificate, Tax receipts, Property photos, Old patta documents converted by OCR

🤖 AI/OCR Pipeline — Auto-read Land Documents

⭐ **ENHANCEMENT (Inspired by EHR Architecture) — This feature is NEW, added to make Land Registration smarter and more secure**

In the EHR project, scanned prescriptions and lab reports are auto-read by AI. We apply the same to old land documents.

Stage	Tool	What It Does for Land Records
Pre-processing	OpenCV	Converts scanned document image to grayscale, removes noise, enhances DPI and contrast for better OCR accuracy
OCR	PyTesseract	Reads text from scanned sale deeds, old survey records, handwritten patta documents, court orders
NLP Extraction	SpaCy + Custom Rules	Identifies: owner name, survey number, land area (acres/cents), district, village, boundaries (N/S/E/W), transaction date, market value, encumbrances
Data Structuring	FastAPI	Converts extracted text into structured land record JSON: { survey_no, owner_name, area, boundaries, district, value, encumbrances }
Confidence Check	Threshold 80%	If OCR confidence < 80% → flag for manual review. Prevents incorrect data from entering blockchain.
FHIR-equivalent	Land Record Schema	Similar to FHIR for health, we use standardized land record JSON schema so any government system can read it

🔐 Consent Management — Controlled Access to Land Records

⭐ **ENHANCEMENT (Inspired by EHR Architecture) — This feature is NEW, added to make Land Registration smarter and more secure**

In the EHR project, patients approve/deny doctor access to health records. We apply the same for land record access by banks, lawyers, and buyers.

Scenario	Who Requests	What Happens
Bank loan processing	Bank officer requests to view land ownership records	Owner gets notification → approves bank access for 7 days → bank views records → access auto-expires
Property sale	Buyer's lawyer requests to verify ownership chain	Owner approves one-time view access → lawyer confirms ownership history → access revoked after view
Legal dispute	Court requests certified land history	Collector org approves court access → full history provided with blockchain TX IDs → immutable evidence
Mortgage check	Government housing scheme verifies ownership	Owner grants read-only access → scheme verifies no encumbrance → access logged and expired

3. Orderer Organization — Members 1, 2, 3

DEDICATED CONSENSUS MANAGERS — These 3 members ONLY manage orderers. They ensure ALL transactions (original + enhanced) are ordered correctly.

M1 Orderer	Member 1 — Orderer 1 Manager (Primary Leader) Primary Raft Leader Orderer (Port 7050)
📦	**Package Transactions into Blocks** Collects all endorsed transactions — land applications, DID registrations, IPFS CID records, consent grants — packages into blocks and delivers to all peers.
🏆	**Raft Leader** Orderer 1 is the elected LEADER in Raft consensus. All transactions come here first. If it crashes, Orderer 2 or 3 automatically takes over.
🆔	**Orders DID Creation Transactions (Enhancement)** ⭐ **ENHANCEMENT** When a new citizen registers and a DID is created, Orderer 1 ensures this DID creation transaction is ordered and committed to blockchain before any land application is allowed.
🌐	**Orders IPFS CID Storage Transactions (Enhancement)** ⭐ **ENHANCEMENT** When documents are uploaded to IPFS and the CID needs to be stored on blockchain, Orderer 1 packages these CID transactions. The CID is immutably tied to the owner's DID on ledger.
🔍	**Monitor Logs** Watches orderer health constantly. If down, restart immediately. Network halts if 2+ orderers fail simultaneously.

M2 Orderer	Member 2 — Orderer 2 Manager (First Backup) Follower Orderer — Auto-failover (Port 8050)
🔄	**Automatic Failover** If Orderer 1 crashes, Orderer 2 participates in Raft election and becomes leader. No land registration transactions — including DID and IPFS CID transactions — are lost.
📋	**Log Replication** Continuously copies the full transaction log from Orderer 1. Keeps ALL transaction types in sync including enhanced features.
🔐	**Consent Transaction Backup (Enhancement)** ⭐ **ENHANCEMENT** Consent grant/revoke transactions (when owner approves bank or buyer access) are replicated here. If Orderer 1 fails during a consent transaction, Orderer 2 ensures it is not lost.

M3 Orderer	Member 3 — Orderer 3 Manager (Second Backup + Channel Admin) Second Backup + Channel Configuration (Port 9050)
💾	**Disaster Recovery** Regular backups of orderer data. If both Orderer 1 and 2 fail, Orderer 3 maintains the network and all land records remain safe.
📡	**Channel Configuration**

	Manages land-registration-channel configuration. Handles policy updates, new member additions, endorsement policy changes.
📊	**Network Health Monitoring** ⭐ **ENHANCEMENT** Monitors blockchain health: block height, TPS, peer connectivity. Also monitors IPFS daemon status since CID transactions depend on IPFS being available.

4. Org1 — Registration Department (Members 4, 5, 6)

Org1 is the FRONT DOOR — Citizens come here. Enhanced with DID generation for land owners, IPFS document storage, DigiLocker verification, and the citizen-facing website.

M4 Org1	**Member 4 — Org1 Admin + CA Manager + DID Setup + DigiLocker Registration** Organization Admin | Fabric CA (Port 7054) | DID Config | DigiLocker Credentials
🏢	**Setup Org1 Identity on Blockchain** Creates MSP folder structure with CA certificates, peer certificates, admin certificates. Generates crypto material for all Org1 members. Runs setOrg1.sh environment script.
🎫	**Run Fabric CA for Org1 (Port 7054)** The Certificate Authority issues X.509 digital ID cards to every Org1 user — peers, admins, citizens, clerks. Every person who uses the system gets a certificate from here.
🆔	**Configure DID Creation System for Land Owners (Enhancement)** ⭐ **ENHANCEMENT** Sets up the DID generation module in chaincode. When a citizen registers, Fabric CA + DID chaincode creates a unique did:hf:org1:<hash> for them. This DID is permanently linked to all their property records.
🔐	**AES-256 Key Management for IPFS (Enhancement)** ⭐ **ENHANCEMENT** Generates and distributes AES-256 encryption keys for each land owner. Documents are encrypted with these keys before IPFS upload. Keys stored in secure Fabric wallet — never on blockchain itself.
🔗	**DigiLocker App Registration** Registers the land registration app on developer.digitallocker.gov.in. Gets CLIENT_ID and CLIENT_SECRET. Sets DIGILOCKER_CLIENT_ID, DIGILOCKER_CLIENT_SECRET, DIGILOCKER_REDIRECT_URI in .env file.
🆔	**Enhancement: DID Generation Config** Configures the chaincode + CA integration that creates a unique DID for each new land owner on registration.
🔗	**DigiLocker: App Credentials** Registers on DigiLocker developer portal, manages CLIENT_ID and CLIENT_SECRET for government document access.

M5 Org1	**Member 5 — Anchor Peer (Peer0, Org1) + DigiLocker OAuth + IPFS Upload API** Anchor + Endorsing Peer (Port 7051) | DigiLocker OAuth Backend | IPFS Integration
🚪	**Official Gateway of Org1** Peer0 is the Anchor Peer. All cross-org communication between Org1, Org2, Org3 goes through this peer. First endorser for all citizen land applications.
✅	**Endorses Land Applications and DID Transactions** Runs createApplication() and createDID() chaincode. Signs endorsements for citizen submissions. Without Org1's endorsement, no application proceeds.
🔗	**DigiLocker OAuth Flow Backend**

	Builds complete OAuth2 flow: /auth/digilocker (redirect to DigiLocker), /auth/digilocker/callback (exchange code for token), /digilocker/documents (list citizen's docs), /digilocker/fetch-document (get specific doc + generate hash).
🌐	**IPFS Upload API for Land Documents (Enhancement)** ⭐ **ENHANCEMENT** Builds API endpoints for IPFS integration: citizen uploads sale deed → API encrypts with AES-256 → uploads to IPFS daemon → receives CID → stores CID on blockchain via storeLandDocumentCID() chaincode. Also handles retrieval: fetch CID from blockchain → fetch file from IPFS → decrypt → serve to authorized user.
📤	**Document Update Flow (Enhancement)** ⭐ **ENHANCEMENT** When land ownership changes (sale/inheritance): new documents uploaded → encrypted → IPFS upload → new CID → updateDocumentCID() chaincode updates the record. Old CID remains accessible for ownership history audit.
📦	**Document Hash Storage** Stores SHA-256 hash of each DigiLocker document on blockchain via storeDocumentHash(). Later verification by Revenue Dept compares stored hash with re-fetched document hash.
🌐	**Enhancement: IPFS Upload/Fetch API** Builds complete IPFS integration — encrypt → upload → store CID on blockchain → fetch by CID → decrypt for authorized access.
🔗	**DigiLocker: Complete OAuth2 Backend** Full OAuth2 "Connect DigiLocker" flow — redirect, callback, token exchange, document fetch from government servers.

M6 Org1	**Member 6 — Committing Peer (Peer2, Org1) + Full Citizen Frontend Developer** Committing Peer (Port 9051) | Next.js Citizen Website | Enhanced UI
💾	**Commit Blocks to Org1 Ledger** Validates and permanently writes all Org1 transactions to blockchain. Checks all 3 org endorsements, verifies policy, writes to CouchDB world state.
🏛️	**Citizen Portal — Core Pages** Builds: Home, Register/Login, Apply for Land Registration (with DigiLocker connect), Track Application Status (timeline view), Download Certificate.
🆔	**DID Display on Citizen Profile (Enhancement)** ⭐ **ENHANCEMENT** Builds citizen profile page showing their DID: "Your Land Owner Identity: did:hf:org1:a3f92bc..." — citizens can copy this DID to share with banks/lawyers for verified ownership proof.
📁	**Document Vault Page (Enhancement)** ⭐ **ENHANCEMENT** Builds "My Documents" vault page: shows all documents stored on IPFS with CID, upload date, document type (Sale Deed, Survey Map, EC). Citizen can view, download, or revoke access to each document.
🔐	**Consent Management UI (Enhancement)** ⭐ **ENHANCEMENT** Builds the Access Requests screen: citizen receives notification when bank/lawyer/buyer requests land record access. Shows: requester name, organization, purpose, requested duration. Citizen taps APPROVE or DENY. Shows active access list with REVOKE button.
📊	**Enhanced Status Timeline** Builds enriched application status page: shows each step with timestamp, officer name, blockchain transaction ID, and link to verify on blockchain explorer.
🔐	**Enhancement: Consent Approval UI**

	Citizen receives access request notifications, approves/denies with one tap. Active consents shown with revoke option.
📁	**Enhancement: IPFS Document Vault UI** Visual document vault showing all land documents stored on IPFS with their CIDs and access history.

5. Org2 — Revenue Department (Members 7, 8, 9)

Org2 is the VERIFICATION TEAM. Enhanced with AI-powered OCR pipeline to auto-read old land documents, IPFS document verification, and DID-based officer authentication.

M7 Org2	**Member 7 — Org2 Admin + CA Manager + Full Backend API Developer** Org2 Admin | Fabric CA (Port 8054) | Express.js API | AES Encryption
🏢	**Setup Org2 Identity** Creates MSP for Revenue Department. Issues X.509 certificates to Revenue Inspectors, MRO, Superintendents with role attributes embedded: role=RevenueInspector:ecert.
🔧	**Build Complete Backend API (Express.js)** Builds entire fabric-api/ server. All REST endpoints: auth (login/register/logout), land (create/get/verify/survey/approve), DigiLocker (connect/callback/documents/fetch).
🔐	**AES-256 Document Decryption for Revenue Review (Enhancement)** ⭐ **ENHANCEMENT** When Revenue officer needs to view a land document stored on IPFS: API fetches CID from blockchain → fetches encrypted file from IPFS → decrypts using owner's AES key (shared with consent) → serves to officer. All without storing decrypted file anywhere.
🆔	**DID-Based Officer Authentication (Enhancement)** ⭐ **ENHANCEMENT** Revenue officers authenticate using both their JWT token AND their officer DID. API verifies: is this DID registered as a Revenue Inspector on blockchain? Prevents unauthorized people from using stolen credentials.
📡	**Blockchain Gateway** Implements Hyperledger Fabric Gateway: loads officer wallet, connects to network, submits transactions (verifyByRevenue, surveyReportUpdate) and queries (getApplication, queryByStatus).
🔐	**Enhancement: AES Decrypt + IPFS Fetch** Revenue officer views land documents: fetch CID from blockchain → fetch from IPFS → decrypt → display. Document never stored on central server.

M8 Org2	**Member 8 — Anchor Peer (Peer0, Org2) + Document Verifier + IPFS Hash Checker** Anchor + Endorsing Peer (Port 9051) | Revenue Verifier | IPFS Integrity Check
🚪	**Official Gateway of Org2** Registered as Org2 Anchor Peer. All cross-org communication for Revenue Department. Syncs with Org1 and Org3 peers via gossip protocol.
✅	**Endorses Revenue Verifications** Runs verifyByRevenue() chaincode. Enforces state transition: only SUBMITTED → REVENUE_VERIFIED. Signs with Org2 endorsement. Also runs surveyReportUpdate(): REVENUE_VERIFIED → SURVEY_COMPLETED.
🔍	**IPFS Document Integrity Check (Enhancement)** ⭐ **ENHANCEMENT** When Revenue officer reviews documents: compares stored blockchain CID with freshly fetched document CID from IPFS. CID mismatch = document on IPFS was tampered. Also verifies DigiLocker hash. Double verification layer.

🤖	**Coordinates AI-processed Document Intake (Enhancement)** ⭐ **ENHANCEMENT** When OCR pipeline (Member 9) extracts data from old scanned documents, this peer endorses the storeOCRExtractedData() transaction that records AI-extracted survey numbers, boundaries, and values on blockchain.
📋	**Consent Verification Before Document Access** ⭐ **ENHANCEMENT** Before serving any land document to an officer, peer calls checkConsent() chaincode to verify the officer/org has valid active consent from the land owner. No consent = no access.
🌐	**Enhancement: IPFS CID Integrity Verification** Double-checks documents haven't been tampered on IPFS by comparing CIDs. Fraud detection layer.
🤖	**Enhancement: Endorses AI-Extracted Data** Signs endorsements for transactions storing OCR/AI extracted land data onto blockchain.

M9 Org2	**Member 9 — Committing Peer (Org2) + Revenue Dashboard + AI/OCR Pipeline Developer** Committing Peer (Port 11051) | Revenue Dashboard | FastAPI AI Service (OCR+NLP)
💾	**Commit Revenue Blocks to Ledger** Validates and commits all Revenue Dept transactions. Checks all 3 org endorsements before writing to Org2's blockchain copy.
📋	**Revenue Officer Dashboard** Builds internal Revenue staff website: pending applications list, document review interface with hash verification status, survey data entry form (area, boundaries, market value, encumbrances).
🤖	**FastAPI AI Document Processing Service (Enhancement — MAJOR)** ⭐ **ENHANCEMENT** Builds a FastAPI microservice (separate from main Node.js API) that converts scanned/uploaded land documents into structured JSON automatically.
👁️	**OCR Module — PyTesseract (Enhancement)** ⭐ **ENHANCEMENT** Pre-processes scanned land documents: grayscale conversion, noise removal, DPI enhancement, thresholding → runs PyTesseract OCR → extracts raw text from old sale deeds, patta books, survey field notes. Confidence threshold: 80% minimum.
🧠	**NLP Pipeline — SpaCy Land Record NER (Enhancement)** ⭐ **ENHANCEMENT** Trains SpaCy NLP with custom land record Named Entity Recognition: extracts survey_number, owner_name, land_area (with unit: acres/cents/guntas), village, district, taluk, boundaries (NESW), market_value, date_of_registration, encumbrances. Custom regex for survey numbers like "Sy.No.123/4A".
📊	**Structured Land Record JSON Output (Enhancement)** ⭐ **ENHANCEMENT** Maps NLP entities to standard land record JSON schema: { survey_no, patta_no, owner_name, owner_did, area, unit, boundaries, village, taluk, district, registration_date, market_value, encumbrances, confidence_score }. Sent to API → merged with application → stored on blockchain.
🤖	**Enhancement: FastAPI OCR+NLP Microservice (Port 8000)** POST /api/ocr/process → receives document image/PDF → returns structured land record JSON. Revenue staff can digitize old paper records instantly.

6. Org3 — Collectorate Department (Members 10, 11, 12)

Org3 is the HIGHEST AUTHORITY. Enhanced with complete smart contract authorship for all new features (DID, IPFS CID, consent), IPFS node management, and AI-assisted decision support for collectors.

M10 Org3	**Member 10 — Org3 Admin + ALL Chaincode Author + IPFS Node Manager** Org3 Admin | Land + Enhanced Chaincode Author | IPFS Node Operator
🏢	**Setup Org3 Identity** Creates MSP for Collectorate. Issues certificates to District Collector and Collector staff with elevated role attributes for final approval authority.
📜	**Write ALL Original Land Registration Chaincode** Authors: createApplication(), verifyByRevenue(), surveyReportUpdate(), approveByCollector(), getApplication(), getAllLandRequest(), getHistory(), queryByStatus().
🆔	**Write DID Chaincode (Enhancement)** ⭐ **ENHANCEMENT** Authors createLandOwnerDID() — generates unique DID for each citizen on registration. Authors verifyDID() — verifies identity before any land operation. Authors transferOwnership() — when land is sold, DID of new owner is linked.
🌐	**Write IPFS CID Chaincode (Enhancement)** ⭐ **ENHANCEMENT** Authors storeLandDocumentCID() — stores IPFS CID on blockchain linked to owner DID. Authors updateDocumentCID() — updates CID when documents change. Authors getLandDocuments() — retrieves all CIDs for a property.
🔐	**Write Consent Chaincode (Enhancement)** ⭐ **ENHANCEMENT** Authors requestLandRecordAccess() — bank/lawyer requests access. Authors grantAccess() — owner approves. Authors revokeAccess() — revoke manually or on expiry. Authors checkConsent() — verify before serving data. Authors getAccessLogs() — full audit trail.
🤖	**Write OCR Data Storage Chaincode (Enhancement)** ⭐ **ENHANCEMENT** Authors storeOCRExtractedData() — stores AI-extracted land data on blockchain with confidence score. Authors flagForManualReview() — marks records where OCR confidence < 80% for human verification.
🌐	**IPFS Node Management** ⭐ **ENHANCEMENT** Runs IPFS daemon (go-ipfs). Manages IPFS node: port 4001 (swarm), 5001 (API), 8080 (gateway). Ensures all land document files persist and are accessible via CID. Sets up IPFS pinning to prevent garbage collection.
🌐	**Enhancement: IPFS Node Operator** Runs the IPFS daemon. All encrypted land documents are stored here. Returns CID for each upload.
📜	**Enhancement: DID + Consent Chaincode Author** Writes ALL smart contracts for DID management, IPFS CID storage, and consent access control.

M11 Org3	**Member 11 — Anchor Peer (Peer0, Org3) + Final Approval + Consent Authority**

	Anchor + Endorsing Peer (Port 11051) | Collector Approval | Consent Enforcement
🚪	**Official Gateway of Org3 — Highest Authority Peer** Most powerful peer in the network. Represents the District Collector's office. All final approvals and consent enforcements go through this peer.
🏆	**Final Endorsement — Land Registration** Endorses approveByCollector() transactions. Without Org3's signature, no land registration is approved. This peer's endorsement is legally mandatory by network policy.
🔐	**Consent Grant/Revoke Authority (Enhancement)** ⭐ **ENHANCEMENT** Runs grantAccess() and revokeAccess() chaincode. When a citizen approves a bank's request to view land records, this peer endorses the consent transaction as the highest authority. Consent is only valid with Org3 endorsement.
🆔	**DID Verification Authority (Enhancement)** ⭐ **ENHANCEMENT** Runs verifyDID() chaincode for all high-stakes operations (ownership transfer, mortgage registration). As the Collector org, Org3 DID verification is the final identity check before any major land transaction.
📢	**Blockchain Events** Emits events: ApplicationApproved (land registration complete), OwnershipTransferred (sale complete), AccessGranted/Revoked (consent events). Frontend listens for real-time notifications to citizens.
📋	**Access Log Recording (Enhancement)** ⭐ **ENHANCEMENT** Every access to land records — whether by bank, lawyer, buyer, or government — is permanently logged: requestor DID, owner DID, timestamp, access type, scope, reason. Immutable audit trail.
🔐	**Enhancement: Consent Smart Contract Executor** All consent grant/revoke chaincode runs through this peer. Owner approval is enforced here.

M12 Org3	**Member 12 — Committing Peer (Org3) + Collector Dashboard + AI Decision Support + DevOps** Committing Peer (Port 13051) | Collector Dashboard | AI Insights | DevOps
💾	**Final Commit — Permanent Registration** This peer's commit is the LAST step. After the District Collector approves, this peer validates all endorsements and writes land registration permanently to blockchain. Immutable forever.
🖥️	**District Collector Dashboard** Builds the Collector approval interface: queue of SURVEY_COMPLETED applications, full review with all documents (fetched from IPFS), Revenue verification report, Survey measurements, Approve/Reject/Request Info buttons.
📄	**Registration Certificate Generator** Generates official PDF certificate: unique registration number, owner DID, property details, blockchain transaction ID, QR code linking to blockchain record, digital signature of Collector.
🤖	**AI Decision Support for Collector (Enhancement)** ⭐ **ENHANCEMENT** Builds an AI insights panel on the Collector dashboard: "This property has 3 previous ownership transfers", "Market value is 15% below district average — verify survey", "Encumbrance certificate shows active loan from 2022 — flag for review". Helps Collector make informed decisions faster.
📊	**Land Record Analytics Dashboard (Enhancement)** ⭐ **ENHANCEMENT**

	Builds analytics: total registrations this month by district/taluk, average processing time per stage, pending applications heatmap by location, OCR success rate, top officers by verification count.
🐳	**DevOps — Full System + IPFS Monitoring** ⭐ **ENHANCEMENT** Manages Docker Compose for entire system: orderers, peers, CAs, CouchDB, IPFS daemon, FastAPI AI service, Node.js API, Next.js frontend. Monitors IPFS storage usage, blockchain height, AI service uptime. Runs setup-network.sh and test-network.sh.
🤖	**Enhancement: AI Decision Support Panel** Collector sees AI insights about each application — anomaly detection, valuation comparison, ownership chain analysis.
🐳	**DevOps: Full System + IPFS Health** Keeps all containers running. Monitors IPFS storage, blockchain health, and AI service availability.

7. Complete Enhanced Transaction Flow

7A. Citizen Registration + DID Creation (NEW STEP)

Step	Who	Action	Result
1	M6 (Frontend)	Citizen opens website, fills: name, Aadhaar, DOB, address	—
4	M4 (CA)	Fabric CA validates and enrolls citizen	X.509 Certificate issued
2	M7 (API)	Registration request sent to blockchain	—
3	M10 (Chaincode)	createLandOwnerDID() invoked — unique DID generated	DID: did:hf:org1:a3f92...
5	M5+M8+M11	All 3 peers endorse DID creation transaction	Endorsing
6	M1 (Orderer)	DID block packaged and delivered	Block Created
7	M6+M9+M12	All committing peers write DID to ledger	DID REGISTERED ✅
8	M6 (Frontend)	Citizen sees their DID on profile page	Ready to Apply

7B. Land Application with DigiLocker + IPFS (ENHANCED)

Step	Who	Action	Result
1	M6 (Frontend)	Citizen clicks "Connect DigiLocker" on application form	—
2	M5 (OAuth)	Redirects to DigiLocker OAuth login page	—
3	Citizen	Logs in with Aadhaar, selects documents on DigiLocker	—
4	M5 (OAuth)	Receives auth code → exchanges for access token	Token Ready
5	M5 (API)	Fetches Aadhaar + Sale Deed + Encumbrance Certificate	Docs Fetched
6	M5 (API)	SHA-256 hash of each DigiLocker doc stored on blockchain	Hashes Stored
7	M5 (API)	Documents encrypted with AES-256 → uploaded to IPFS	⭐ NEW
8	M10 (IPFS)	IPFS daemon returns unique CID for each document	CIDs Ready
9	M10 (Chaincode)	storeLandDocumentCID() — CIDs linked to owner DID on blockchain	⭐ NEW — CIDs on chain
10	M7 (API)	Land application submitted with DID + CIDs + hashes	—
11	M5+M8+M11	All 3 peers endorse createApplication() transaction	Endorsing
12	M1 (Orderer)	Application block packaged	Block Created
13	M6+M9+M12	All committing peers write to ledger	SUBMITTED ✅

7C. OCR — Old Paper Documents Digitized (NEW FEATURE ⭐)

Step	Who	Action	Result

1	Revenue Officer	Scans old physical patta/sale deed document	Image/PDF ready
2	M9 (FastAPI)	Image pre-processed: grayscale, noise removal, DPI enhance	Clean Image
3	M9 (OCR)	PyTesseract reads text from document	Raw text extracted
4	M9 (NLP)	SpaCy NER extracts: survey_no, owner, area, boundaries, value	Entities Found
5	M9 (FastAPI)	Structured JSON created with confidence score	JSON + Score
6	Check	Confidence >= 80%? → proceed / < 80%? → flag for manual review	Quality Check
7	M7 (API)	storeOCRExtractedData() called with JSON + confidence score	—
8	M8 (Peer)	Peer endorses OCR data storage transaction	Endorsing
9	M5+M8+M11	All 3 orgs endorse	—
10	M9 (Committer)	Commits OCR data to blockchain — old record now digital	DIGITIZED ✅

7D. Revenue Verification + IPFS Document Check (ENHANCED)

Step	Who	Action	Result
1	M8 (Dashboard)	Revenue officer opens application in dashboard	—
2	M7 (API)	checkConsent() — verifies officer org has access (auto-approved for Revenue dept)	Access OK
3	M7 (API)	Fetches document CIDs from blockchain	CIDs Retrieved
4	M5 (API)	Fetches encrypted files from IPFS using CIDs	Files Fetched
5	M7 (API)	Decrypts files with AES key → displays to officer	Docs Visible
6	M8 (Peer)	Verifies DigiLocker hash AND IPFS CID both match	⭐ Double Verify
7	Officer	Reviews documents, checks land details, fills verification form	—
8	M7 (API)	verifyByRevenue() submitted with officer DID signature	—
9	M5+M8+M11	All 3 orgs endorse	Endorsing
10	M9 (Committer)	Commits verification to blockchain	REVENUE_VERIFIED ✅

7E. Collector Approval + Certificate (ENHANCED)

Step	Who	Action	Result
1	M12 (Dashboard)	Collector opens SURVEY_COMPLETED application	—
2	M12 (AI Panel)	AI insights shown: ownership history, value analysis, risk flags	⭐ AI Support

3	M12 (Dashboard)	Collector reviews all: docs (from IPFS), revenue report, survey	—
4	M7 (API)	approveByCollector() submitted — M11 gives final Org3 endorsement	—
5	M5+M8+M11	All 3 orgs endorse final approval	Endorsing
6	M1 (Orderer)	Final block created	Block Created
7	M12 (Committer)	FINAL COMMIT — permanently on blockchain	APPROVED ✅
8	M11 (Event)	ApplicationApproved event emitted	Notification Sent
9	M12 (Certificate)	PDF certificate generated: reg number + owner DID + blockchain TX ID + QR code	Certificate Ready
10	M6 (Frontend)	Citizen receives notification and downloads certificate	COMPLETE 🎉

7F. Bank/Lawyer Accesses Land Records — Consent Flow (NEW ⭐)

Step	Who	Action	Result
1	Bank Officer	Logs into bank portal, searches citizen by DID or Aadhaar	—
2	M7 (API)	requestLandRecordAccess() called: bank DID, owner DID, purpose="Loan Verification", duration=7 days	Request Sent
3	M8 (Peer)	Endorses access request transaction	On Blockchain
4	M6 (Frontend)	Citizen gets notification: "HDFC Bank requests access to your land records for 7 days"	—
5	Citizen	Reviews request and taps APPROVE	—
6	M11 (Peer)	grantAccess() endorsed by Org3 (consent authority)	Consent Recorded
7	M5+M8+M11	All 3 orgs endorse consent grant	CONSENT ACTIVE ✅
8	Bank	Bank officer now views land records in their portal	—
9	M7 (API)	Every view: checkConsent() verifies → fetch CID → fetch IPFS → decrypt → serve	Access Logged
10	M11 (Peer)	Each access logged: bank DID, timestamp, action, scope	Audit Trail
11	Auto	After 7 days: revokeAccess() chaincode auto-triggers	EXPIRED ⏰
12	Citizen	Can manually revoke anytime from Access Requests page	REVOKED 🔒

8. Complete Chaincode Functions (Original + Enhanced)

Function	Type	Who Triggers	What It Does	Status Change
initLedger()	Original	Deploy time	Initializes blockchain with empty state	None
createApplication()	Original	Citizen submits	Saves land application with DID + CIDs attached	SUBMITTED
verifyByRevenue()	Original	Revenue officer	Records revenue verification with officer DID	REVENUE_VERIFIED
surveyReportUpdate()	Original	Inspector	Saves survey data: area, boundaries, value	SURVEY_COMPLETED
approveByCollector()	Original	Collector	Final approval + registration number generation	APPROVED
getApplication()	Original	Any query	Returns single application by ID	Read only
getAllLandRequest()	Original	Dashboard	Returns all applications filtered by role	Read only
getHistory()	Original	Audit view	Full timeline of all changes with TX IDs	Read only
queryByStatus()	Original	Filter view	Returns applications with given status	Read only
storeDocumentHash()	Original	Doc upload	Saves SHA-256 DigiLocker doc hash on chain	None
verifyDocumentHash()	Original	Revenue verify	Compares stored vs current doc hash	Returns bool
createLandOwnerDID()	⭐ Enhanced	User register	Generates unique DID for land owner	DID Created
verifyDID()	⭐ Enhanced	Before any op	Verifies identity via DID before critical actions	Returns bool
transferOwnership()	⭐ Enhanced	Property sale	Links new owner DID to property, preserves history	TRANSFERRED
storeLandDocumentCID()	⭐ Enhanced	Doc to IPFS	Stores IPFS CID linked to owner DID on blockchain	CID Stored
updateDocumentCID()	⭐ Enhanced	Doc updated	Updates CID when document changes, keeps old CID in history	CID Updated
getLandDocuments()	⭐ Enhanced	View docs	Returns all CIDs for a property	Read only
storeOCRExtractedData()	⭐ Enhanced	AI pipeline	Stores OCR-extracted land data with confidence score	OCR Stored
flagForManualReview()	⭐ Enhanced	Low confidence	Marks records where OCR < 80% for human check	FLAGGED
requestLandRecordAccess()	⭐ Enhanced	Bank/Lawyer	Creates access request on blockchain	Request Pending

grantAccess()	⭐ Enhanced	Owner approves	Records consent with scope + expiry on chain	Access Active
revokeAccess()	⭐ Enhanced	Owner/Auto-expiry	Revokes active consent	Access Revoked
checkConsent()	⭐ Enhanced	Before serving data	Verifies requester has valid active consent	Returns bool
getAccessLogs()	⭐ Enhanced	Audit	Returns full log of all accesses to a property	Read only

9. Quick Reference — All 12 Members

#	Member	Org	Blockchain Role	Original Job	⭐ Enhanced Job
1	Orderer 1 Manager	Orderer	Primary Raft Leader	Block creation + delivery	Orders DID + IPFS CID transactions
2	Orderer 2 Manager	Orderer	Follower / Failover	Backup for network	Replicates consent transactions
3	Orderer 3 Manager	Orderer	Backup + Channel Admin	Channel configuration	Monitors IPFS daemon status
4	Org1 Admin	Org1	CA + MSP Manager	DigiLocker credentials	DID system config + AES key management
5	Org1 Anchor Peer	Org1	Anchor + Endorsing Peer	DigiLocker OAuth backend	IPFS upload/fetch API integration
6	Org1 Commit Peer	Org1	Committing Peer	Citizen website (Next.js)	DID profile + Document vault + Consent UI
7	Org2 Admin	Org2	CA + MSP Manager	Express.js Backend API	AES decrypt + DID officer auth
8	Org2 Anchor Peer	Org2	Anchor + Endorsing Peer	Revenue verification + survey	IPFS CID integrity check + OCR data endorser
9	Org2 Commit Peer	Org2	Committing Peer	Revenue dashboard	FastAPI AI: OCR+NLP pipeline
10	Org3 Admin	Org3	CA + MSP Manager	Land chaincode author	DID+IPFS+Consent chaincode + IPFS node
11	Org3 Anchor Peer	Org3	Anchor + Endorsing Peer	Collector final approval	Consent enforcement + DID authority
12	Org3 Commit Peer	Org3	Committing Peer	Collector dashboard + DevOps	AI decision support + analytics + DevOps

Port & Service Reference

Service	Port	Manager	Purpose
Orderer 1	7050	Member 1	Primary ordering — all transactions
Orderer 2	8050	Member 2	Backup orderer
Orderer 3	9050	Member 3	Second backup + channel admin
Fabric CA Org1	7054	Member 4	Citizen enrollment + DID generation
Peer0.Org1 (Anchor)	7051	Member 5	Land + DigiLocker + IPFS endorsement
Peer2.Org1 (Commit)	9051	Member 6	Citizen-side final commit
Fabric CA Org2	8054	Member 7	Revenue staff enrollment
Peer0.Org2 (Anchor)	9051	Member 8	Revenue + IPFS verification endorsement
Peer2.Org2 (Commit)	11051	Member 9	Revenue-side final commit
Fabric CA Org3	9054	Member 10	Collector + consent service enrollment
Peer0.Org3 (Anchor)	11051	Member 11	Collector approval + consent authority
Peer2.Org3 (Commit)	13051	Member 12	Final commit — both original + enhanced

Backend API (Express.js)	3001	Member 7	All land registration + consent APIs
FastAPI AI Service ⭐	8000	Member 9	OCR + NLP document processing
Frontend (Next.js)	3000	Member 6	Citizen portal + consent UI
IPFS API ⭐	5001	Member 10	Land document upload/fetch
IPFS Gateway ⭐	8080	Member 10	Document read access
IPFS Swarm ⭐	4001	Member 10	IPFS peer-to-peer network
CouchDB	5984	Member 12	World state queries

10. Setup Order — Who Does What First

Phase	Step	Who	Action
Phase 1Crypto	1	M4,M7,M10	cryptogen generate --config=crypto-config.yaml
Phase 1Crypto	2	M4	docker-compose up ca.org1.example.com (port 7054)
Phase 1Crypto	3	M7	docker-compose up ca.org2.example.com (port 8054)
Phase 1Crypto	4	M10	docker-compose up ca.org3.example.com (port 9054)
Phase 2Network	5	M1,M2,M3	docker-compose up orderer1 orderer2 orderer3
Phase 2Network	6	M5,M8,M11	Start anchor peers: peer0.org1, peer0.org2, peer0.org3
Phase 2Network	7	M6,M9,M12	Start committing peers: peer2.org1, peer2.org2, peer2.org3
Phase 3Channel	8	M3	peer channel create -c land-registration-channel
Phase 3Channel	9	All peers	peer channel join -b land-registration-channel.block
Phase 3Channel	10	M5,M8,M11	peer channel update -f OrgMSPanchors.tx (set anchor peers)
Phase 4IPFS ⭐	11	M10	ipfs init && ipfs daemon (ports 4001, 5001, 8080)
Phase 5Chaincode	12	M10	Package chaincode: peer lifecycle chaincode package land-reg.tar.gz
Phase 5Chaincode	13	M5,M8,M11	Install: peer lifecycle chaincode install land-reg.tar.gz
Phase 5Chaincode	14	M4,M7,M10	Approve: peer lifecycle chaincode approveformyorg
Phase 5Chaincode	15	M10	Commit: peer lifecycle chaincode commit
Phase 6Services	16	M9 ⭐	Start FastAPI AI: uvicorn main:app --host 0.0.0.0 --port 8000
Phase 6Services	17	M7	Start Backend API: cd fabric-api && npm install && npm start
Phase 6Services	18	M4	Configure .env: DIGILOCKER credentials + AES_KEY + IPFS_API_URL
Phase 6Services	19	M6	Start Frontend: cd client && npm install && npm run dev
Phase 7Test	20	M12	./setup-network.sh && ./test-network.sh

11. Beginner Glossary — All Terms

Term	⭐	Simple Explanation
Blockchain		Database stored in linked blocks. Once written, cannot be changed. Like permanent ink.
Hyperledger Fabric		Private blockchain for businesses. Only authorized members join — unlike public Bitcoin.
Chaincode / Smart Contract		Code on blockchain that runs automatically when triggered. Like a vending machine rule.
Orderer		Server that sorts transactions in correct order and packages into blocks. Like a post office.
Raft Consensus		3+ orderers vote to agree on order. Need 2 of 3 to continue. Network survives 1 crash.
Anchor Peer		Official representative of an org to other orgs. Main cross-org contact point.
Endorsing Peer		Peer that simulates a transaction and signs it as valid before it goes to orderer.
Committing Peer		Peer that validates and permanently writes final blocks to the ledger.
MSP		Membership Service Provider — manages who belongs to the org via digital certificates.
Fabric CA		Certificate Authority — issues X.509 digital ID cards to all members.
DigiLocker		Government of India platform storing official documents. 100% verified — cannot be faked.
SHA-256 Hash		Mathematical fingerprint of a document. 1 character change = completely different hash.
JWT		JSON Web Token — secure login token. Proves identity without resending password each time.
Docker		Runs all services in isolated containers that work the same on any computer.
CouchDB		Database for blockchain world state. Enables fast rich queries on current data.
TLS		Transport Layer Security — encrypts all network communication.
DID	⭐	Decentralized Identifier — permanent blockchain identity for land owner. did:hf:org1:<hash>. Cannot be duplicated or forged.
IPFS	⭐	InterPlanetary File System — global decentralized storage. Files get a permanent CID address.
CID	⭐	Content Identifier — unique fingerprint/address of a file on IPFS. Stored on blockchain instead of full file.
AES-256	⭐	Military-grade encryption for land documents before IPFS upload. Only owner's key decrypts.
OCR	⭐	Optical Character Recognition — reads text from scanned images. Used for old paper land records.
PyTesseract	⭐	Python OCR library. Reads text from scanned sale deeds, patta books, survey field notes.
NLP	⭐	Natural Language Processing — AI understands text. Identifies survey numbers, owner names, land areas.

SpaCy	⭐	Python NLP library with Named Entity Recognition (NER) trained for land record terminology.
FastAPI	⭐	Python API framework for the AI microservice (OCR+NLP). Separate from main Node.js backend.
Consent Management	⭐	Smart contract system where land owner must approve every bank/lawyer/buyer access request.
Access Log	⭐	Permanent record on blockchain of every time someone accessed land records — who, when, why.
NER	⭐	Named Entity Recognition — AI that identifies specific entities in text like names, numbers, locations.
OAuth 2.0		Login protocol. "Connect DigiLocker" uses this — like "Login with Google" on websites.
Endorsement Policy		Rule: ALL 3 orgs must sign every transaction. No single org can approve alone.